



Straightforward synthesis of [(2*S*,4*R*)-1-cyclohexyl-4-methylpiperidin-2-yl]-methanol and [(2*S*,4*R*)-1-cyclohexyl-4-methylpiperidin-2-yl](diphenyl) methanol: novel chiral ligands for the catalytic addition of diethylzinc to benzaldehyde to give rise to an extensive turn in the sense of asymmetric induction

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Dedicated to Professor Antonio García Martínez on his 70th birthday

ABSTRACT

Enantiopure [(2*S*,4*R*)-1-cyclohexyl-4-methylpiperidin-2-yl]methanol **5a** and [(2*S*,4*R*)-1-cyclohexyl-4-methylpiperidin-2-yl]-(diphenyl)methanol **5b**, new β -amino alcohols based on L-pipecolic acid (homoproline), have been prepared straightforwardly from *rac*-alaninol and *rac*-2-amino-1,1-diphenylpropan-1-ol, respectively. The described route constitutes as a model procedure for the preparation of other related C(4) or/and C(3)-substituted 2-piperidinylmethanols.

The new chiral ligands have shown a singular behaviour on the stereocontrol of the benchmark reaction of benzaldehyde and diethylzinc compared with other C(4)-unsubstituted analogues prepared by ourselves from L-pipecolic acid (compounds **5c**, **5d**, **5e** and **5f**). The catalytic activity, the sense of asymmetric induction and the degree of the enantioselectivity depend on the appropriate combination of the substituents on the structural scaffold, but also on the metal-alkoxide involved in the catalysis (zinc or lithium alkoxides). The enantioselective addition of diethylzinc to benzaldehyde mediated by ligands **5a** and **5c** has been studied with DFT methods. The theoretical evaluation was performed in connection with a working hypothesis based on the different loadings of *cis*- and *trans*-catalysts in the reaction medium.

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1. Introduction

The addition of organozinc compounds to prochiral aldehydes in the presence of chiral β -amino alcohols constitutes a highly efficient procedure for the enantioselective construction of C–C bonds.¹ Chiral ligands for asymmetric catalysis have been generally derived from a small number of readily available natural products.^{2–7}

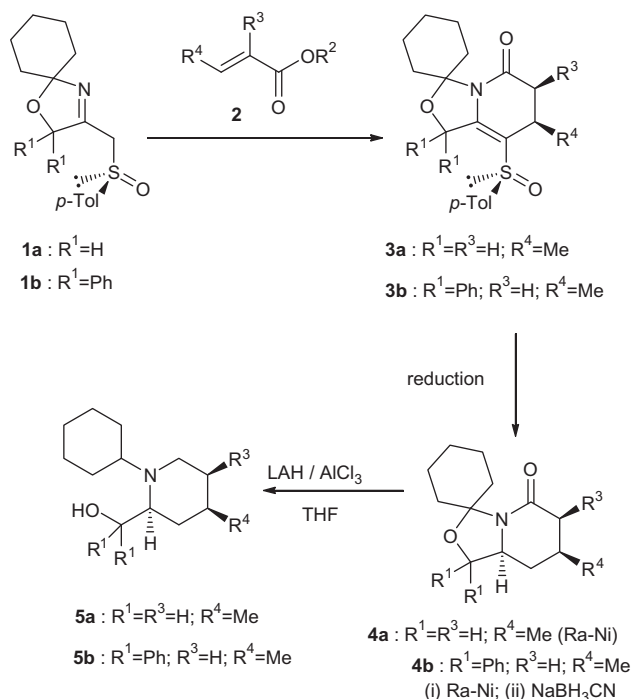
In order to obtain the two enantiomeric products it is necessary to have the two enantiomeric ligands but, in many cases, this is not possible because only one of them is available. Moreover, modulation of the catalytic properties through structural modifications in origin could be limited by the nature of precursors. Thus, the enantioselective synthesis of ligands, with either a different configuration of the stereocentres responsible for the asymmetric induction, or with key substituents to fine tune the catalytic properties, can represent an advantageous alternative to ligands of natural origin and can allow the circumvention of the aforementioned limitation.⁸

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We have recently reported a *one-pot* procedure for the diastereoselective synthesis of (3,4)-substituted 5,6-dehydropiperidin-2-ones such as **3a** (Scheme 1) from α -sulfinyl ketimine **1a**, which is raised from *rac*-alaninol and alkyl acrylate **2**.⁹ In addition, the reaction with *Ra*-Ni and reduction of the acetal-amide intermediate with alane would result with the isolation of a sole amino alcohol enantiomer **5a**. This strategy has important advantages: (a) Although the chiral auxiliary, (–)-menthyl *p*-toluenesulfonate, is sacrificed on the synthesis, this is easily prepared from (–)-menthol and sodium *p*-toluenesulfonate on a scale of 50 g. (b) A library of lactams **3** can be prepared by modifying the nature of the starting *rac*-amino alcohol and the alkyl acrylate, and also by changing the chirality of the menthyl sulfinate. (c) The amino alcohols **5** are obtained with full control of all structural and stereochemical parameters. (d) This synthetic methodology generates up to three stereocentres through the chiral control of reagent **1** in the convergent step of the synthesis.

These advantages encouraged us to consider that properly substituted pipecolic acid-based ligands would constitute an interesting class of chiral catalysts by comparison with other ones such as 2-aziridinyl¹⁰ and 2-azetidinylmethanols,¹¹ and prolinols,¹² which have been largely evaluated in the alkylation reaction



Scheme 1.

of aldehydes with organozinc reagents. The synthesis and evaluation of 2-piperidinylmethanols have not been investigated, due to the dual (*cis-trans*) stereochemistry of the bicyclic zinc-chelated complexes formed from these β -amino alcohols, which are the actual catalytic species.¹³

Although the competitive transition states generated from the coexistence of the two diastereomeric catalysts could result in a low enantioselectivity, it can be used as a tool to gain information on the role of the substituents in the ring and also in the side chain on the stereochemical outcome of the reaction. The work described herein details our efforts in this area.

2. Results and discussion

2.1. Synthesis of β -amino alcohols

The syntheses of amino alcohols **5a** and **5b** (Chart 1) were carried out from the chiral α -sulfinyl ketimines **1a** and **1b**,

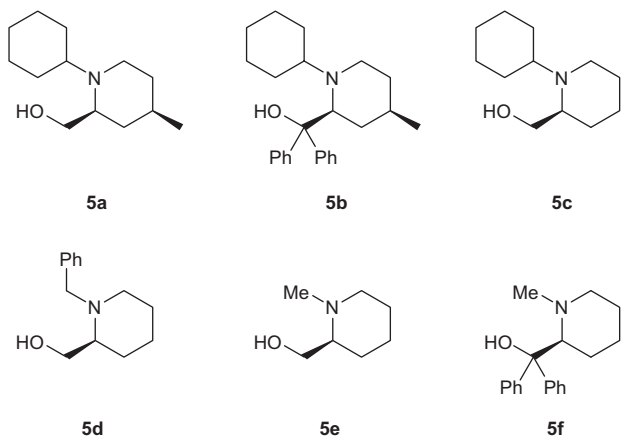


Chart 1.

respectively, through 5,6-dehydropiperidin-2-ones intermediates (Scheme 1). Compound **1b** was obtained from *rac*-2-amino-1,1-diphenylpropan-1-ol in 60% yield, following the procedure described for **1a** and other analogues.⁹ Compound **1b** was then transformed into lactam **3b** via the addition of (*E*)-methyl crotonate, mediated by butyllithium. Monitoring the reaction by TLC shows a single adduct, with its evolution to the cyclic product being complete within 16 h. The crude was purified by flash LC and the structure of the isolated product (75% yield) was established as **3b** by ¹H and ¹³C NMR. The absolute configuration of the new stereocentre (C-4) was assigned as (*R*) by comparison with **3a**.⁹ In this case, the key signal became that of the methyl group in the piperidone ring [δ 0.42 ppm for the (*R*)-configured diastereomer and δ 1.31 ppm for the (*S*)-configured one]. Compound **3b** showed a signal for the methyl group at δ 0.29 ppm. Furthermore the stereochemical course of the reaction would be also invoked in order to establish an identical configuration for **3a** and **3b**.

Next, **3b** was derivatised as the lactam, via desulfinylation Raney-Ni in EtOH/benzene at 60 °C. Despite the harsh reaction conditions being explored, the enamine proved reluctant to react and the product of the hydrogenation **4b** could not be detected in the reaction crude. Thus, the enamine-lactam was isolated and treated with $NaCNBH_3$ /TFA in acetic acid for 16 h to give a product in 67% yield, which was identified by NMR as **4b**, the major isomer in a mixture (86:14) of two diastereomers. Attempts to separate both stereoisomers were unsuccessful. Thus, the mixture was treated with a THF solution of alane, generated in situ from $LiAlH_4$ and $AlCl_3$ (3:1), for 16 h at room temperature. After a careful work-up, compound **5b** was obtained as the only isomer in 58% yield. Its structure and configuration were determined from the NMR spectroscopic data.

On the other hand, compound **5a** was obtained by an analogous route from **3a**, following the strategy depicted in Scheme 1. In this case, precursor **4a** was easily derived from **3a** by hydrogenation with Ra-Ni in EtOH at 60 °C for 20 min in a *one-pot* procedure (quantitative yield), and its reduction with alane provided the amino alcohol **5a** (70% yield).

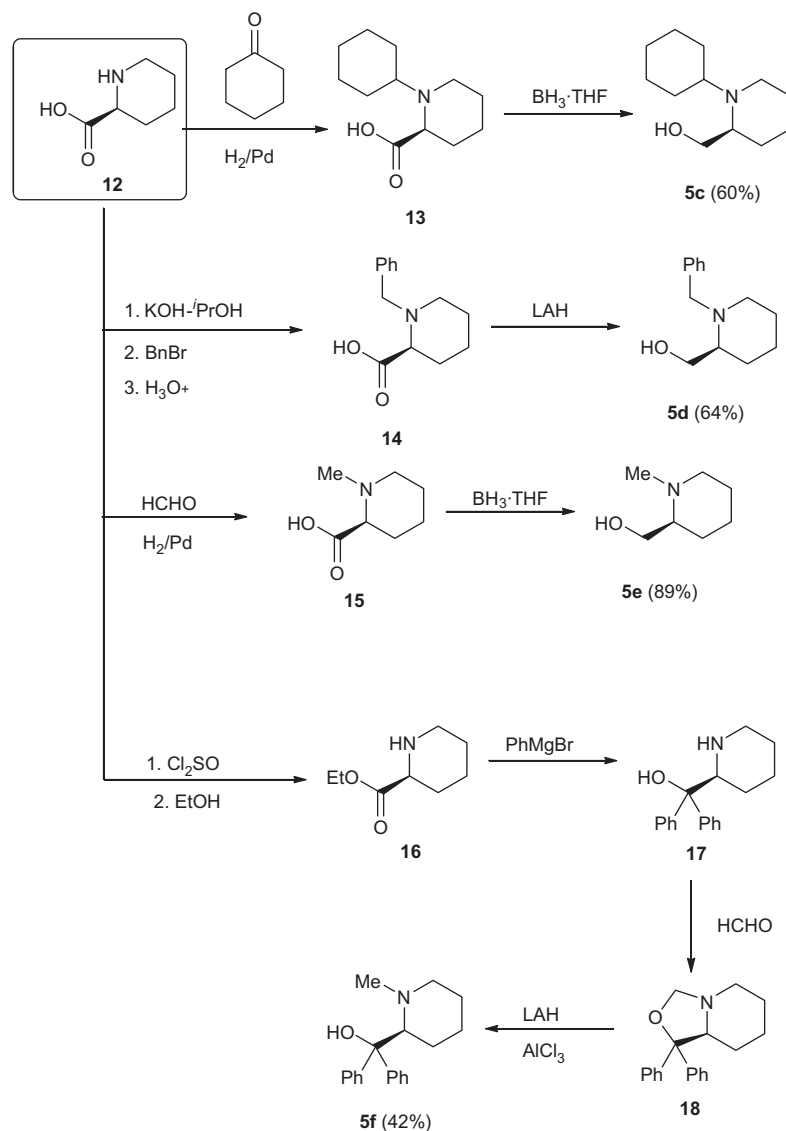
As expected, the configuration of the new stereocentre in the C-2 became (*S*), taking into account the stereocontrol induced by the substituent on C-4 of the enamine-lactam precursor allowing attack of the reagent to be *anti*. The (*S*)-configuration is consistent with the axial character of the hydrogen atom on C-2, together with the values observed for the vicinal coupling constants ³ $J_{H2,H3}$ and ³ $J_{H2,H3'}$ in **5a** (³ $J_{H2,H3} = 11.4$ Hz and ³ $J_{H2,H3'} = 3.2$ Hz) and also in **5b** (³ $J_{H2,H3} = 11.4$ Hz and ³ $J_{H2,H3'} = 3.8$ Hz).

In order to compare the catalytic activity and enantioselectivity of ligands **5a** and **5b** with other N-substituted (2*S*)-piperidinylmethanols, we obtained the β -amino alcohols **5c–5f** from the commercially available L-pipecolinic acid (Chart 1), following the reaction sequences outlined in Scheme 2.

Overall yields in the range 42–89% were reached by application of standard procedures in order to allow access to the ligands **5c–5f**.

2.2. Enantioselective addition of diethylzinc to benzaldehyde catalysed by 2-piperidinylmethanols

Our first aim was to evaluate the catalytic activity and the degree of enantioselection shown by the chiral ligands **5a–5f** in the standard reaction of benzaldehyde and diethylzinc. Both parameters should be key probes in order to gain a better insight as to the structure of competitive transition states. Furthermore, as the structure is dependent on the nature of the organometallic species which is the actual catalyst, we carried out two sets of experiments. In the first, we generated a zinc chelate in the reaction medium from the diethylzinc reagent and a catalytic amount of the ligand. In the second experiment, an organolithium complex



Scheme 2.

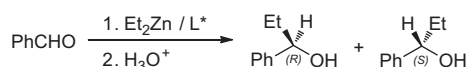
was generated by adding a stoichiometric amount of *n*-butyllithium to the ligand, before the reagent and benzaldehyde were added. The results are shown in Table 1. All reactions were carried out at room temperature using an $\text{Et}_2\text{Zn}/\text{C}_6\text{H}_5\text{CHO}$ /ligand molar ratio of 1.8:1:0.05 in toluene as solvent. As can be seen from Table 1, enantioselectivities ranging from 82% (*R*) for the ligand 5a (entry 2) to 83% (*S*) for the ligand 5b (entry 4) were observed. With the other ligands, a broad range of (*R*)-stereoselection ranging from 60% (*R*) (entry 6) to 2% (*R*) (entry 12) was achieved by modifying the substituent on the nitrogen atom of the piperidine ring, but also introducing a *gem*-diphenyl group in the side chain (entries 11 and 12). This group has been named as the *magic* group in catalyst design and has been frequently used in recent years in order to improve stereoselection.¹⁴

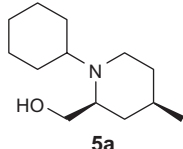
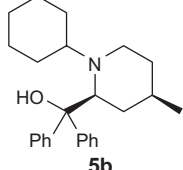
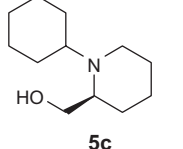
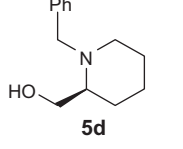
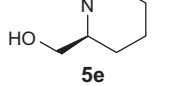
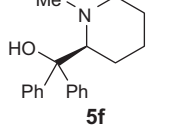
As can be seen in Table 1, the *magic* character of the *gem*-diphenyl group is more apparent for ligand 5b (entries 3 and 4) than for 5f (entries 11 and 12). Nevertheless, the same trend was observed in both cases in terms of the comparison with their analogous 5a (entries 1 and 2) and 5e (entries 9 and 10), respectively. Thus, it seems that the *gem*-diphenyl group enhances the *S*-stereoselection or *like*-induction.^{15,16} In fact, these results are in agreement with those observed for other conformationally restricted heterocyclic

ligands bearing the *gem*-diphenyl group on the side chain (compounds 6, 8, 10 and 11, Fig. 1).

It is important to note that other *gem*-dialkyl groups such as dimethyl and diethyl in the *N*-benzylprolinol (compounds 7a and 7b, Fig. 1)^{12a} do not cause the same effect on the asymmetric induction. On the other hand, the extent of the change observed in going from *N*-methylprolinol 9⁵ to its derivative 8 is more important than that observed for compound 5f from 5e, but less than that observed for 5b from 5a, both bearing the methyl group at the 4-position of the piperidine ring and the bulkier cyclohexyl substituent on the nitrogen atom.

In spite of the catalytic activity, which was evaluated after 1 or 2 h of reaction, all reactions were generally completed within 18 h (Table 1). The stereocontrol depends on the nature of the metal and the substitution on the ring and the side chain, as we have already mentioned. The reaction when controlled by ligands 5a, 5b and 5c, displayed a slight increase in the stereoselectivity when lithium instead of zinc complexes were involved (Table 1, entries 1–6), but an opposite effect was observed (*unlike*-induction) in the reaction controlled by ligands 5d, 5e and 5f (Table 1, entries 7–12). Furthermore, the lithium alkoxides that arose from these ligands became more active than the zinc ones. This finding was also observed for

Table 1The enantioselective addition of Et₂Zn to benzaldehyde using L-pipecolinic acid-based ligands^{a,b}

Entry	Ligand (L [*])	Chiral cat. (L [*] – M)	% yield ^c	ee (config.) ^d	Conv. (%) ^e
1	 5a	5a -Zn	98	76 (R)	—/84
2		5a -Li	98	82 (R)	46/71
3	 5b	5b -Zn	98	79 (S)	60/—
4		5b -Li	98	83 (S)	84/—
5	 5c	5c -Zn	98	57 (R)	79/93
6		5c -Li	98	60 (R)	45/70
7	 5d	5d -Zn	57	53 (R)	26/35
8		5d -Li	98	27 (R)	50/69
9	 5e	5e -Zn	80	53 (R)	16/22
10		5e -Li	90	23 (R)	17/28
11	 5f	5f -Zn	77	11 (R)	23/31
12		5f -Li	98	2 (R)	50/68

^a Reactions were run in toluene at room temperature using Et₂Zn/C₆H₅CHO/L^{*} molar ratio of 1.8:1:0.05 (0.125 M solution of C₆H₅CHO).^b All experiments were performed three times to ensure reproducibility.^c Determined by GC (Cyclodex) at 18 h of reaction (yields: varied by ±2).^d Analysis of the samples at different reaction times showed that ee to be unchanged within the standard limits (enantiomeric excesses: varied by ±2).^e Determined by the standard procedure at 1 h/2 h of reaction in all cases (conversions: varied by ±2).

ligand **5b** (Table 1, entries 3 and 4) since its catalytic activity as lithium alkoxide increased at the same time the stereocontrol (*like-induction*) did.

Ligands **5a** and **5c** show a different trend. Despite their degree of (R)-stereocontrol, the change of metal in the chelate complex from zinc to lithium led to a decrease in the chemical conversion (Table 1, entries 1, 2 and 5, 6). Consequently, all these findings show that the ligands submitted to the comparison bring about very different geometries or relative configurations in the transition states responsible for the stereocontrol in going from the zinc to the lithium alkoxide.

2.3. DFT mechanistic analysis

In order to examine the origin of the enantioselectivity in the reaction of benzaldehyde and diethylzinc mediated by 2-piperidinylmethanols, the reaction mechanism involving ligands **5a** and **5c** was investigated theoretically by means of DFT methods follow-

ing the guidelines proposed by various research groups and using the transition state structure analysis.¹⁷

In Figure 2, we have shown the steps that afford the (R)-configured 1-phenyl-1-propanol, which arises from the *cis*-chelate complex *cis-cat* via a sole transition state (**B**), which has been named *cis-transoid-anti-R* (its geometry is based on Noyori's 5/4/4 model¹³). This notation reports the relative configuration of the substituents at the atoms belonging to the fusion between nuclei of the 6/5/4/4 tetracyclic system [H/R¹: *cis*; R¹/Et(Zn1): *transoid*; Et(Zn1)/Et(Zn2): *anti* and, in addition, the prochirality of the carbonyl group: (R) or (S)]. Another competitive transition state arising from the *trans*-chelate complex named *trans-transoid-anti-S* is also outlined in Figure 2. The mechanism proposed by Noyori is applied here in a simplified scheme, which includes the *product-forming mixed complex* or a *catalyst-reagents complex A* involved in the turnover-limiting step affording the *catalyst-product complex (cis-R)*.

Up to 2⁴ diastereomeric transition states with a 6/5/4/4 geometry of type **B** could be considered, due to the presence of four

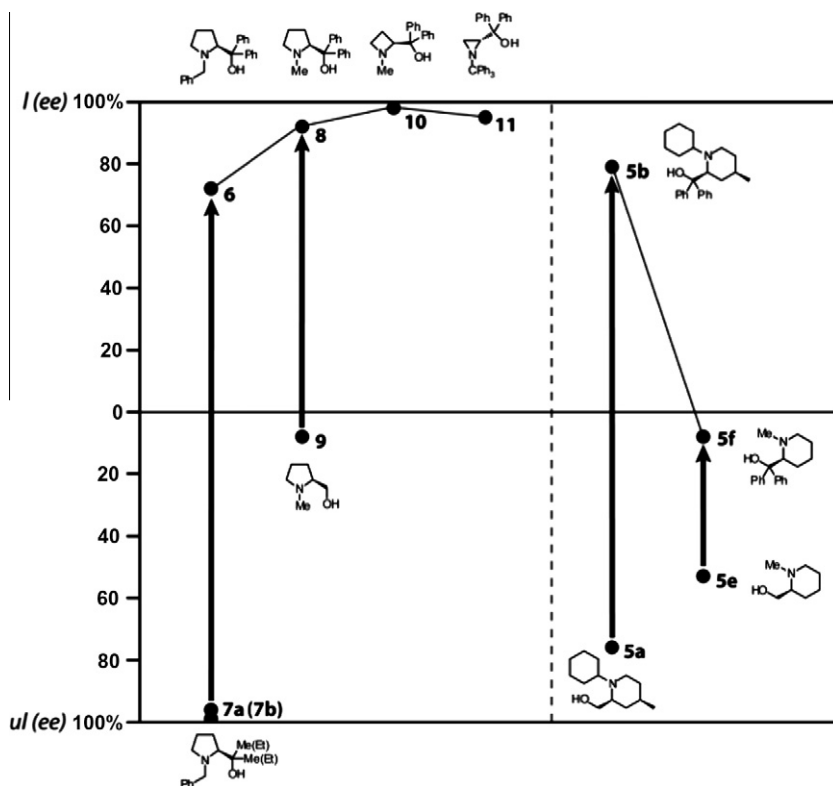


Figure 1. Analogies observed in the effect of the configuration and the substitution on the side chain of 1,2-dialkylsubstituted heterocyclic ligands on the stereocontrol of the reaction of benzaldehyde and diethylzinc.

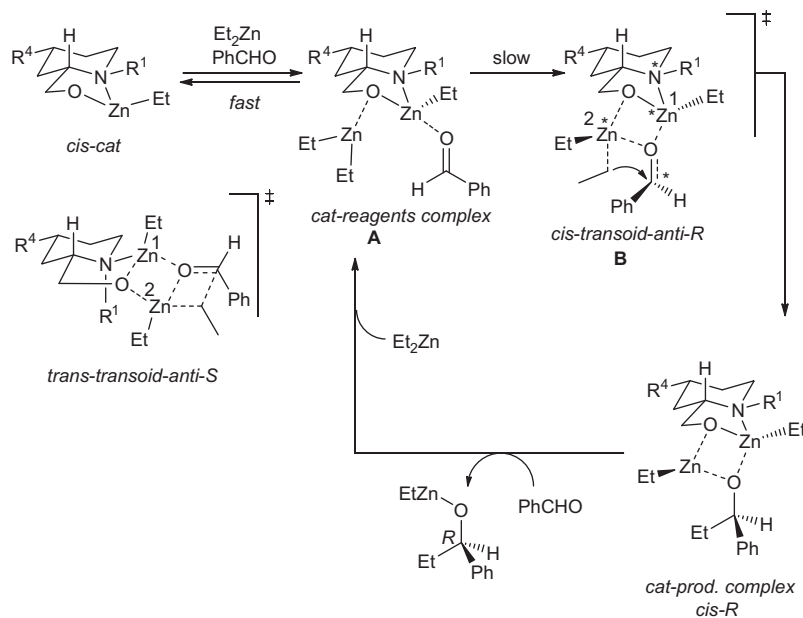


Figure 2. Hypothetical reaction pathways of diethylzinc and benzaldehyde involving *cis*- and *trans*-chelate complexes (*cis*- or *trans*-cat) for (2*S*)-piperidinylmethanols, based on Noyori's outcomes.^{13,17a}

new stereocentres. However, only the four of them have an *anti*-configuration between the non-transferred ethyl groups (and an *E*-relationship between the phenyl group and the coordinative O–Zn1 bond) were chosen (*cis*-*transoid-anti-R*, *cis-cisoid-anti-S*, *trans-transoid-anti-S* and *trans-cisoid-anti-R*). This simplification was checked with other cases.^{8,17b,17g}

At this point, it should be noted that the *cis*/*trans*-configuration is pertinent to the acid-base reaction between the ligand and diethyl-

zinc and, as a consequence, two catalytic species are generated non-reversibly, being different in both catalytic efficiency and concentration or actual load (Fig. 3). Hence, a statistical factor weighting the different loading of each catalyst in the reaction medium should be introduced in Eq. (1) that reports the final stereoselectivity [N_R/N_S is the molar fraction of the (*R*)- and (*S*)-1-phenyl-1-propanol] as $\exp - (\Delta G_{cis-cat}^\ddagger - \Delta G_{trans-cat}^\ddagger) / RT$, accounting for stereocontrol in the step, which leads to the *cis*- and *trans*-catalysts (Fig. 3).

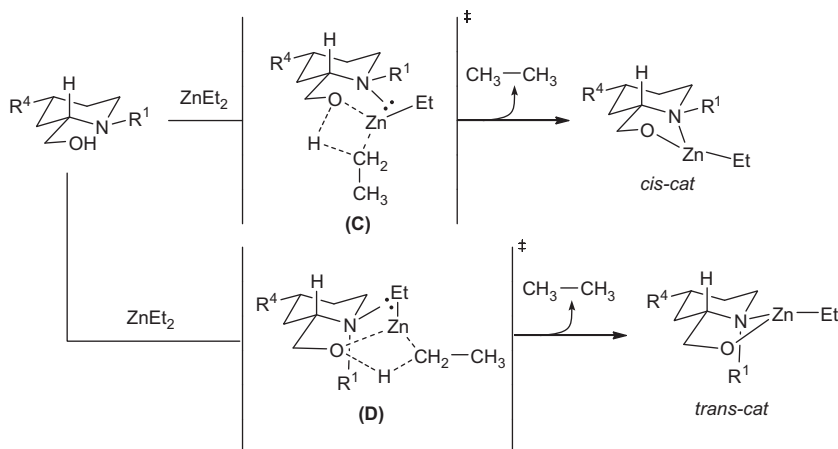


Figure 3. Competitive acid-base reactions between ligands **5** and diethylzinc affording *cis* and *trans* catalytic species.

On the other hand, the second factor in Eq. (1), $\exp - (\Delta\Delta G_{\text{cis}(R)}^{\ddagger} - \Delta\Delta G_{\text{trans}(S)}^{\ddagger})/RT$, considers a single competitive reaction pathway at each turnover limiting step giving rise to a single enantiomer [*R*] from the *cis*-cat and [*S*] from *trans*-cat, Figure 2]. Thus, an evaluation of the relative efficiency of the catalysts requires the energetic differences between each transition state and the appropriate ground state of the same configuration ($\Delta\Delta G^{\ddagger}$) to be calculated. In doing so, the reaction pathway which matches the lowest energy transition state and the least stable product-forming mixed complex will be the most efficient in the migration of the ethyl group to the incoming carbonyl one.

$$\frac{N_R}{N_S} = e^{-(\Delta G_{\text{cis-cat}}^{\ddagger} - \Delta G_{\text{trans-cat}}^{\ddagger})/RT} \cdot e^{-(\Delta\Delta G_{\text{cis}(R)}^{\ddagger} - \Delta\Delta G_{\text{trans}(S)}^{\ddagger})/RT} \quad (1)$$

Despite the dual *cisoid/transoid* configuration of the two product-forming mixed complexes (Fig. 2), a mechanism of dissociation and recombination could be assumed,^{17a} allowing an equilibrium to be established to stabilise the diastereomeric complexes through each catalytic species and the reagents. Hence, a dynamic kinetic asymmetric transformation (DYKAT)¹⁸ would occur, providing the two enantiomeric alkoxide-products in one preferential bias (*unlike*- or *like*-induction) from each catalyst [*cis*(*R*) and *trans*(*S*), respectively].

In order to check these working hypotheses, the geometries and relative energies of the structures **A** and **B** (Fig. 2), and **C** and **D** (Fig. 3) were calculated in the first approach by using the ground state and transition state searching algorithms implemented in GAUSSIAN 03.¹⁹ The geometries were preoptimised by a PM3 semi-empirical method²⁰ and the final structures were optimised at

the B3LYP²¹/6-31G(d)²² (for C, H, O and N) and LanL2DZ²³ (for Zn) levels. All the final structures of transition states showed a single imaginary frequency corresponding to either the transfer of the hydrogen atom at the hydroxyl group of the ligand to diethylzinc (in the in situ formation of the catalyst) or the migration of the suitably oriented ethyl group to the carbonyl carbon during the rate limiting step (Fig. 2).

On the other hand, non imaginary frequencies were found in the analysis of optimised structures of product-forming mixed complexes **A** (Fig. 2). Both the geometries and energy levels are summarized in the Tables 2–4 for ligands **5a** and **5c**.

The characterisations of the transition states affording each catalyst were performed and the results are reported in Table 2. The ring inversion of the piperidine ligands has three reaction pathways that should be considered. Thus, ligand **5a** becomes more stereoselective than **5c**. This seems to be due to the methyl group at C4 of the piperidine ring, which prevents all axial conformers being competitive (Table 2, entry 2). A different behaviour was observed with ligand **5c**, since the 1,2-diaxial conformer (Table 2, entry 5) became the most reactive in terms of the energy level of its transition state. This is possible because the antiperiplanar relationship of the two substituents prevents their mutual interaction.

Thus, a statistical factor weighing the different loading of *cis*- and *trans*-catalysts of 22.8 has been calculated for ligand **5a** and other one of 6.3 for **5c** $\{\exp - (\Delta G_{\text{cis-cat}}^{\ddagger} - \Delta G_{\text{trans-cat}}^{\ddagger})\}/RT$ in Eq. (1).

The relative energies calculated for the optimised geometries of transition states corresponding to the migration of ethyl group from the product-forming mixed complexes of type **A** (Fig. 2) are summarised in Tables 3 and 4 for ligands **5a** and **5c**, respectively.

Table 2

Geometries and DFT B3LYP/6-31G(d):LanL2DZ relative energies (kcal/mol) calculated for transition states giving to the *cis*- and *trans*-catalyst for ligands **5a** and **5c**

Entry	Ligand	TS (type) ^a	TS geometry ^b	E_{RHF}^c /a.u.	ν_i (cm ⁻¹)	ZPE ^d	Relative energies ^e (kcal/mol)	N_{cis} or N_{trans}^f
1	5a	C (<i>cis</i>)	<i>eq,eq,eq</i>	−864.398445	994.7i	0.499374	0	95.8
2	5a	C (<i>cis</i>)	<i>ax,ax,ax</i>	−864.390980	1011.1i	0.500012	4.68	n.p. ^g
3	5a	D (<i>trans</i>)	<i>ax,eq,eq</i>	−864.395496	1087.7i	0.499175	1.85	4.2
4	5c	C (<i>cis</i>)	<i>eq,eq</i>	−825.081098	1049.3i	0.471407	0.80	17.8
5	5c	C (<i>cis</i>)	<i>ax,ax</i>	−825.082371	1008.5i	0.471661	0	68.5
6	5c	D (<i>trans</i>)	<i>ax,eq</i>	−825.080838	1069.6i	0.471282	0.96	13.7

^a H/Chex (*cis* or *trans*) as described for *cis*-cat and *trans*-cat in Figure 3.

^b Conformational isomerism related to the substituents on the nitrogen, C(2) and C(4) atoms at the piperidine ring, respectively.

^c Optimised geometries at the B3LYP/6-31G(d) (for C, H, O and N) and LanL2DZ (for Zn) level.

^d Unscaled zero-point correction.

^e All energies are relative to the most stable transition state.

^f Calculated stereoselectivity assuming a Boltzmann distribution at room temperature.

^g Non-participating way.

Table 3

Geometries and DFT B3LYP/6-31G(d);LanL2DZ relative energies (kcal/mol) of ground and transition states with respect to the most stable ground state for ligand **5a**

Structure ^a	$E_{\text{RHF}}^{\text{b}}$ /a.u.	ν_i (cm ⁻¹)	ZPE ^c	Relative energies (kcal/mol)
<i>cis-cisoid</i>	–1354.240231		0.668618	5.26
<i>cis-transoid</i>	–1354.248616		0.667776	0.00
<i>trans-cisoid</i>	–1354.236102		0.668816	7.85
<i>trans-transoid</i>	–1354.241854		0.668044	4.24
<i>cis-cisoid-anti-S</i>	–1354.218736	195.1i	0.670000	18.75
<i>cis-transoid-anti-R</i>	–1354.235035	222.6i	0.669229	8.52
<i>trans-cisoid-anti-R</i>	–1354.218209	237.6i	0.669273	19.08
<i>trans-transoid-anti-S</i>	–1354.229121	229.6i	0.669764	12.23

^a H/Chex (*cis* or *trans*), Chex-Et(Zn1) (*cisoid* or *transoid*) and Et(Zn1)/Et(Zn2) *anti* as described for *cis(trans)-cat* and TS corresponding to Figure 2.
^{b,c,d} See Table 2 for optimised geometries, unscaled zero-point correction and relative energies.

Table 4

Geometries and DFT B3LYP/6-31G(d);LanL2DZ relative energies (kcal/mol) of ground and transition states with respect to the most stable ground state for ligand **5c**

Structure ^a	$E_{\text{RHF}}^{\text{a}}$ /a.u.	ν_i (cm ⁻¹)	ZPE ^a	Relative energies (kcal/mol)
<i>cis-cisoid</i>	–1314.931699		0.640993	1.04
<i>cis-transoid</i>	–1314.933905		0.639883	0.00
<i>trans-cisoid</i>	–1314.928942		0.640643	3.11
<i>trans-transoid</i>	–1314.927233		0.640298	4.19
<i>cis-cisoid-anti-S</i>	–1314.911050	194.4i	0.641540	14.34
<i>cis-transoid-anti-R</i>	–1314.920122	218.4i	0.640930	8.65
<i>trans-cisoid-anti-R</i>	–1314.909090	234.1i	0.641472	15.57
<i>trans-transoid-anti-S</i>	–1314.914364	228.9i	0.640582	12.26

^a See Table 3.

Table 5

Catalytic efficiency of chelate–complexes involved in the turnover limiting steps affording *cis* and *trans*-catalyst–product complexes for the reaction of diethylzinc and benzaldehyde promoted by ligands **5a** and **5c**

L [*]	Structure	$\Delta\Delta G^\ddagger$ (kcal mol ⁻¹) ^a	Relative energy ^b	$\exp(-\Delta\Delta G^\ddagger/RT)$ (N_R/N_S) ^c /%
5a	<i>cis-cisoid-anti-S</i>	13.48	5.50	
5a	<i>cis-transoid-anti-R</i>	8.52	0.53	0.41 (29.1)
5a	<i>trans-cisoid-anti-R</i>	11.23	3.24	
5a	<i>trans-transoid-anti-S</i>	7.99	0.00	1.00 (70.9)
5c	<i>cis-cisoid-anti-S</i>	12.96	4.89	
5c	<i>cis-transoid-anti-R</i>	8.65	0.58	0.38 (27.5)
5c	<i>trans-cisoid-anti-R</i>	12.46	4.39	
5c	<i>trans-transoid-anti-S</i>	8.07	0.00	1.00 (72.5)

^a Energy differences between each transition state and the ground state of the same configuration (calculated from the relative energies summarised in the Tables 3 and 4).

^b Relative energies with respect to lowest value of $\Delta\Delta G^\ddagger$.

^c Calculated stereoselectivity on the turnover limiting steps.

The zero-point energy values (ZPE) and imaginary frequencies have been included for their characterisation. The correlative geometries of precursors such as **A**, *cis-cisoid(transoid)* and *trans-cisoid(transoid)* have also been calculated in each case. Hence, only two turnover limiting steps to each *cis(R)* and *trans(S)* catalyst-product complexes became competitive.

The similar results observed for both reactions mediated by ligands **5a** and **5c** are more apparent from the calculation of $\Delta\Delta G^\ddagger$ values in Eq. (1), as can be seen in Table 5. It can be seen from the values obtained for the catalytic efficiency of *cis*- and *trans*-catalysts performed from **5a** and **5c** (0.41 and 0.38, respectively), that the *trans-transoid* product-forming mixed complex is more efficient than the *cis-transoid* one for each ligand, although non-remarkable differences between ligands were found. By introducing each factor and their exponential statistical ones in Eq. (1), the stereoselectivity could be calculated as $(N_R/N_S)_{5a} = 9.36$ and $(N_R/N_S)_{5c} = 2.39$ which implies enantiomeric excesses up to 80.8% (*R*) for **5a** and 41% (*R*) for **5c**. These results are in good agreement with the observed values (76% and 57%, respectively). In addition,

this indicates that the stereochemical outcome of the reaction is mainly determined by the stereoselection provided in preforming the *cis*- and *trans*-catalyst.

Inspection of the four structures drawn in the Figure 4 disclosed that the most preorganised ground state affording the *trans-transoid-anti-S* transition state is the most competitive pathway. It is consistent with the data about the non-bonding distance between the carbon atom of carbonyl group and the ethyl radical which is considerably foreshortened in the *trans-transoid* ground state with respect to the *cis-transoid* one (3.779 Å for **5a** and 3.457 Å for **5c** vs 3.821 Å for **5a** and 3.779 Å for **5c**, respectively).

All geometric parameters at the 4/4 core of the hypothetical structures are in agreement with those calculated by Yamakawa and Noyori.^{17k} This analysis provides the correct sense and extension of the stereoselectivity, and in addition, it explains the modulation caused by the substituent on C4 at the piperidine ring on the stereocontrol. An extensive experimental and theoretical study is currently in progress in order to clarify other structural effects on the basis of the DYKAT checked herein.

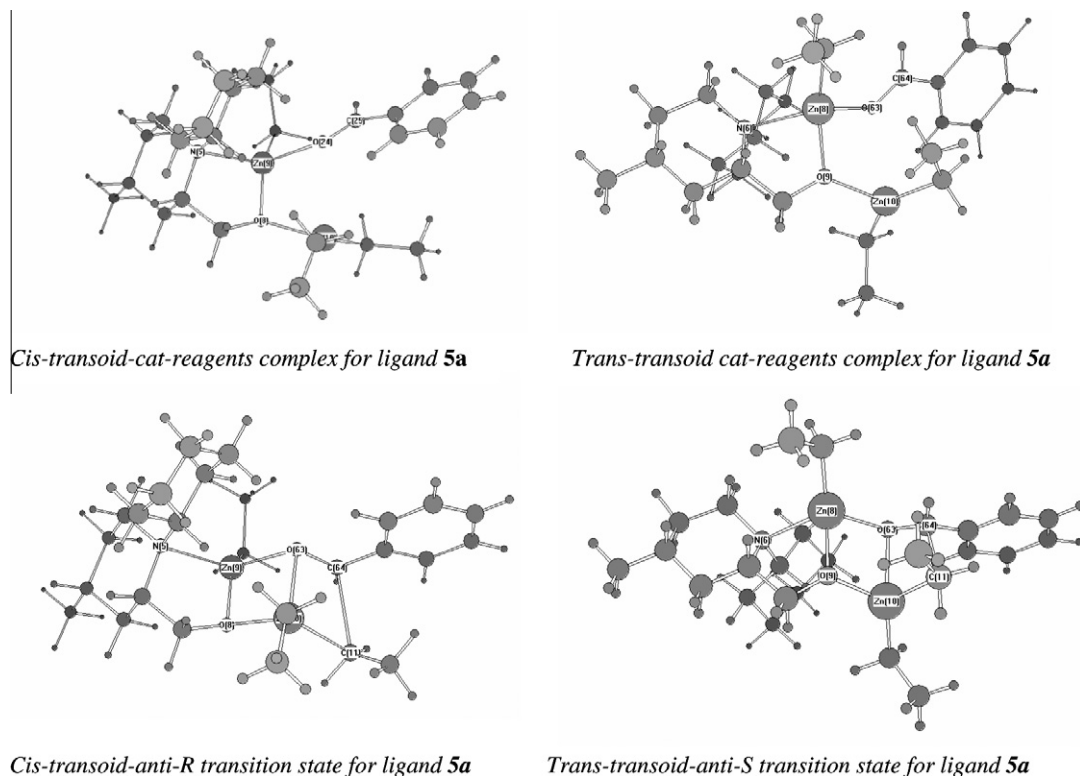


Figure 4. Significant calculated structures for *cat*-reagent complexes and transition states of ligand **5a**.

3. Conclusions

In conclusion, enantiopure [(2*S*,4*R*)-1-cyclohexyl-4-methylpiperidin-2-yl]methanol **5a** and [(2*S*,4*R*)-1-cyclohexyl-4-methylpiperidin-2-yl](diphenyl)methanol **5b** have been straightforwardly prepared from *rac*-alaninol and *rac*-2-amino-1,1-diphenylpropan-1-ol, respectively.

These chiral ligands show a singular behaviour on the stereocontrol of the benchmark reaction of benzaldehyde and diethylzinc compared with other piperidylmethanols **5c**, **5d**, **5e** and **5f** prepared by ourselves from *L*-pipecolinic acid, but also with other previously reported on the related proline.

Thus, ligand **5a** gave the highest stereoselection (*unlike* induction), up to 82% (*R*), while **5b** switched the enantiomeric excess up to 83% (*S*) (*like* induction). The 4-unsubstituted ligands **5c–5f** showed a weak stereoselection [60–2% ee of (*R*)-1-phenyl-1-propanol] which could be modulated not only by modifying the size of the substituent on the nitrogen atom (Chex > Bn > Me) but also by introducing the *gem*-diphenyl group on the sidechain. Furthermore, the use of the most reactive lithium alkoxides prepared from these latter ligands provided a lowest (*R*)-stereoselection due to the fact that those transition states affording (*S*)-1-phenyl-1-propanol are relatively stabilised when Li-catalysts are used.

Theoretical studies based on Noyori's 5/4/4 model and DFT methods have allowed us to justify the effect of the substituents at C4 of the piperidine ring of ligands **5a** and **5c** on the stereocontrol of the benchmark reaction. It has been supported in a working hypothesis proposing that the two catalysts (*cis* and *trans*) are formed in the reaction medium, each one giving rise to two product-forming mixed complexes to be in fast equilibrium with the catalyst and the reagents. Among of all the turnover limiting steps leading to (*R*)- and (*S*)-configured products, only one from each catalyst, *cis-R* and *trans-S*, became efficient (*trans-S* more than *cis-R*). Despite their relative catalytic activity, a statistical factor related to the actual loading of each catalyst, is responsible for

the final stereoselectivity. Further experiments and theoretical calculations are now in progress in order to study other structural effects on the stereochemical outcomes of the reaction.

4. Experimental section

4.1. Materials and methods

The ^1H and ^{13}C NMR spectra were recorded at 300 or 500 MHz and 75 or 125 MHz, in a Bruker AC-300 or Bruker AM-500 spectrometer, respectively, using CDCl_3 . The chemical shifts (δ) refer to TMS (^1H) or deuterated chloroform (^{13}C) signals. Coupling constants (J) are reported in Hertz. Multiplicities in the proton spectra are indicated as s (singlet), d (doublet), t (triplet), q (quartet), quint. (quintuplet) and m (multiplet). Elemental analyses were performed with a Perkin–Elmer 2400 C, H, N analyzer. Optical rotations were measured at room temperature (20–23 °C) using a Perkin–Elmer 241 MC polarimeter (concentration in g/100 mL). Infrared spectra were recorded on a Shimadzu FTIR-8300 spectrophotometer or a Bruker Tensor 27 spectrophotometer. The HRMS (EI) were performed on a Bruker APEX Qe 4.7 T spectrometer by electrospray ionisation (ESI). The enantiomeric excesses were determined by GC (Hewlett–Packard 5890) using a capillary column Cyclodex-B and nitrogen as carrier gas.

All reactions in non-aqueous media were carried out in a flame-dried glassware under an argon atmosphere. Reagents and solvents were handled by using standard cannulae or syringe techniques. Tetrahydrofuran (THF) and diethyl ether were distilled from sodium and benzophenone, dichloromethane from P_2O_5 and isopropanol from CaH_2 . In all other cases, commercially available reagent-grade solvents were employed without purification. Analytical TLC was routinely used to monitor reactions. Plates pre-coated with Merck Silica Gel 60 F254 of 0.25 mm thickness were used, and visualised with UV light, with either anisaldehyde/sulfuric acid/ethanol (2:1:100) or phosphomolybdic acid solution (PMA,

10% in ethanol) as developing agents. Merck Silica Gel 60 (230–400 ASTM mesh) was employed for flash column chromatography. Chemicals for reactions were used as purchased from the Aldrich Chemical Co.

4.2. Theoretical calculations

DFT calculations were performed using the GAUSSIAN 03 software package with the option TS of Opt keyword to request optimisations to a transition state.¹⁹ All structures were optimised using density functional theory and the non-local hybrid Becke's three-parameters exchange functional (denoted as B3LYP)²¹ with LanL2DZ pseudopotential and the associated basis set for Zn,²³ and the 6-31G(d)²² basis set for C, H, O and N atoms. All transition states and local minima were validated through frequency analysis.

4.3. Synthesis of 5-(*p*-tolylsulfinyl)-5,6-dehydro-piperidin-2-ones **3a** and **3b**. Typical procedure

To a cold (−78 °C) solution of sulfinyl ketimine (+)-**1a** or (+)-**1b** (3.17 mmol) in dry THF (60 mL) under an argon atmosphere was added a solution of *n*-BuLi (1.6 M in hexane, 2.3 mL, 3.68 mmol). The solution was stirred at −78 °C for 0.5 h, after which the temperature was raised to −30 °C and methyl (*E*)-2-butenate (6.33 mmol, 0.15 M solution in THF) was added. The reaction mixture was then kept at rt and stirred for 3 h. The mixture was hydrolysed with a saturated aqueous NH₄Cl solution (25 mL) and extracted with EtAcO (3–30 mL). The combined organic layers were washed with brine, dried over Na₂SO₄, concentrated under reduced pressure and chromatographed (silica gel; hexane/EtAcO: 1:1; v/v).

4.3.1. (−)-(S₅,7'R)-7'-Methyl-8'-[(4-methylphenyl)sulfinyl]-6',7'-dihydrospiro[cyclohexane-1,3'-[1,3]oxazolo[3,4-*a*]pyridin]-5'(1'H)-one, (−)-**3a**

From (+)-**1a** (927 mg, 3.17 mmol) and methyl (*E*)-2-butenate (6.33 mmol, 0.15 M solution in THF) following the procedure described above. Two fractions of 695 mg and 353 mg of **3a** and its C(7')-epimer, respectively, were obtained in 92% yield. Both isomers were identified through NMR spectroscopic data by comparison with samples previously prepared.

4.3.2. (−)-(S₅,7'R)-7'-Methyl-8'-[(4-methylphenyl)sulfinyl]-1',1'-diphenyl-6',7'-dihydrospiro[cyclohexane-1,3'-[1,3]oxazolo[3,4-*a*]pyridin]-5'(1'H)-one (−)-**3b**

From (+)-**1b** (413 mg, 0.931 mmol) and methyl (*E*)-2-butenate (1.85 mmol, 0.15 M solution in THF) following the procedure described above. The reaction mixture was stirred at rt for 16 h and the crude was purified by LC to obtain 375 mg of (−)-**3b** (75% yield). White solid; mp 210–211 (decomp.); [α]_D = −367.0 (c 1.0, CHCl₃); IR (ATR) 1689, 1646, 1369, 1319, 1030 cm^{−1}; ¹H NMR (300 MHz, CDCl₃) δ 0.29 (d, 3H, *J* = 7.0 Hz, CHCH₃), 0.87 (d, 1H, ²*J* = 12.8 Hz, cyclohexyl), 1.16–1.45 (m, 2H, cyclohexyl), 1.50 (dt, 1H, ²*J* = ³*J* = 13.0 Hz, ³*J* = 3.5 Hz, cyclohexyl), 1.57–1.93 (m, 4H, cyclohexyl), 2.13 (td, 1H, ²*J* = ³*J* = 13.0 Hz, ³*J* = 4.9 Hz, cyclohexyl), 2.36 (dd, 1H, ²*J* = 16.1 Hz, ³*J* = 1.9 Hz, COCH₂), 2.32 (s, 3H, ArCH₃), 2.64 (m, 1H, cyclohexyl), 2.79 (dd, 1H, ²*J* = 16.1 Hz, ³*J* = 6.2 Hz, COCH₂), 3.05 (cdd, 1H, ³*J* = 7.0, 6.2 Hz, 1.9 Hz, CHCH₃), 6.54 (AA'XX' system, 2H, ³*J* = 8.2 Hz, ArH_{ortho}-CH₃), 7.07 (AA'XX' system, 2H, ³*J* = 8.2 Hz, ArH_{ortho}-SO), 7.75–7.79 (m, 2H, ArH), 7.31–7.54 (m, 8H, ArH); ¹³C NMR (75 MHz, CDCl₃) δ 18.0 (CHCH₃), 21.2, 21.2 (ArCH₃, CHCH₃), 22.7, 22.7, 24.3, 31.7, 36.1, 41.7 (COCH₂, cyclohexyl), 89.2 (CO), 98.2 (OCN), 120.3 (C=O), 124.4, 127.8, 127.9, 128.6, 128.7, 128.7, 129.4, 130.1 (ArCH), 137.0, 140.8, 141.2, 144.6, 145.7 (C=CN, ArC), 166.9 (C=O). Anal. Calcd for C₃₂H₃₃NO₃S: C, 75.11; H, 6.50; N, 2.74. Found: C, 75.20; H, 6.57; N, 2.68.

4.4. Synthesis of (7'S, 8a'S)-7'-methyltetrahydrospiro[cyclohexane-1,3'-[1,3]oxazolo[3,4a]pyridin]-5'(1'H)-one, **4a**

To a stirred solution of (−)-**3a** (779 mg, 1.85 mmol) in absolute EtOH (20 mL) was added Raney Nickel W-2²⁴ (2.4 g; 4 mL) and the reaction mixture was heated at 60 °C for 20 min. Then, it was cooled down to rt, filtered on a thin pad of Celite and the metallic salts were washed with ether (5 × 5 mL). The combined organic extracts were evaporated at reduced pressure to give a colourless solid that was purified by flash chromatography (silica gel; hexane/EtAcO: 4:1; v/v) obtaining 470 mg (91% yield) of **4a**. White solid; mp 102–103 °C; [α]_D = +24.0 (c 1.1, CHCl₃); IR (ATR) 1645 cm^{−1}; ¹H NMR (300 MHz, CDCl₃) δ 1.02 (d, 3H, ³*J* = 6.2 Hz, CH₃), 0.99–1.08 (m, 1H, CH₂CHCH₃), 1.20–1.70 (m, 8H, cyclohexyl), 1.87–2.00 (m, 3H, COCH₂, CHCH₃, CH₂CHCH₃), 2.40–2.60 (m, 3H, COCH₂, cyclohexyl), 3.40 (dd, 1H, ²*J* = 8.3 Hz, ³*J* = 10.3 Hz, CH₂O), 3.65 (dddd, 1H, ³*J* = 11.1, 10.3, 5.4, 3.0 Hz, CHN), 4.04 (dd, 1H, ³*J* = 8.3 Hz, ³*J* = 5.4 Hz, CH₂O), ¹³C NMR (75 MHz, CDCl₃) δ 21.5 (CH₃), 23.0, 23.1, 24.5 (cyclohexyl), 28.4 (CHCH₃), 30.9 (cyclohexyl), 33.9 (CH₂CHN), 34.3 (cyclohexyl), 41.1 (COCH₂), 57.4 (CHN), 69.1 (CH₂O), 95.9 (C ipso), 166.8 (C=O). Anal. Calcd for C₁₃H₂₁NO₂: C, 69.92; H, 9.48; N, 6.27. Found: C, 69.94; H, 9.51; N, 6.23.

4.5. Synthesis of (7'S,8a'S)-7'-methyl-1,1'-diphenyltetrahydrospiro[cyclohexane-1,3'[1,3]oxazolo[3,4a]pyridin]-5'(1'H)-one, **4b**

From (−)-**3b** (130 mg, 0.25 mmol) in EtOH (2 mL) and benzene (1 mL), and Raney-Nickel (1.2 g; 2 mL) at 50 °C during 16 h, following the procedure described for **4a**. A colourless solid was isolated in 91% yield (86 mg), which upon flash chromatography (silica gel; hexane/EtAcO: 4:1; v/v) gave an intermediate desulfinylated **4b-enamide**: white solid; mp 166–168 °C; [α]_D = +82.8 (c 1.0, CHCl₃); IR (ATR) 1678, 1448, 1340 cm^{−1}; ¹H NMR (300 MHz, CDCl₃) δ 0.77 (d, 3H, ³*J* = 5.5 Hz, CH₃), 1.22–2.06 (m, 8H, cyclohexyl), 2.11–2.34 (m, 2H, COCH₂, CHCH₃), 2.50 (m, 1H, COCH₂), 2.82 (td, 1H, ²*J* = ³*J* = 13.4 Hz, ³*J* = 4.6 Hz, cyclohexyl), 2.93 (td, 1H, ²*J* = ³*J* = 13.4 Hz, ³*J* = 3.7 Hz, cyclohexyl), 4.58 (d, 1H, ³*J* = 2.4 Hz, CH=C), 7.11–7.69 (m, 10H, ArH); ¹³C NMR (75 MHz, CDCl₃) δ 20.2 (CH₃), 23.2, 23.3, 24.9 (cyclohexyl), 27.4 (CH), 34.1, 35.0 (cyclohexyl), 41.2 (COCH₂), 87.3, 97.4 (CO, OCN), 105.8 (CH=C), 125.5, 125.6, 125.9, 126.0, 126.3, 128.5 (ArCH), 140.9, 144.4, 144.6 (ArC, CH=C), 165.8 (C=O). Anal. Calcd for C₂₅H₂₇NO₂: C, 80.40; H, 7.29; N, 3.75. Found: C, 80.45; H, 7.22; N, 3.79.

To a solution of 70 mg of the desulfinylated enamine (0.19 mmol) in AcOH (4 mL) at rt, 5 μ L of TFA (0.065 mmol) and 30 mg of NaCNBH₃ (0.48 mmol) were added under argon. After stirring for 16 h, a 3 M solution of NaOH was added dropwise until neutral pH. The reaction mixture was then extracted with DCM (3 × 10 mL) and the combined extracts were washed with brine, dried over Na₂SO₄, filtered and concentrated at reduced pressure. The residue was purified by flash chromatography (silica gel; toluene/MeOH: 40:1; v/v) to obtain 47 mg of an inseparable mixture (86:14) of epimers of **4b** (67% yield). IR (ATR) 1660, 1447, 1407, 1046, 1022, 742, 701 cm^{−1}; ¹H NMR (300 MHz, CDCl₃) δ 0.96 (d, 3H, ³*J* = 6.2 Hz, CH₃), 1.26–2.10 (m, 13H, cyclohexyl, COCH₂, CHCH₂CH), 2.38 (ddd, 1H, ²*J* = 16.3 Hz, ³*J* = 4.7, ⁴*J* = 1.7 Hz, CHCH₂CH), 3.05 (td, 1H, ²*J* = 13.1 Hz, ³*J* = 5.0 Hz, COCH₂), 4.41 (dd, 1H, ³*J* = 11.8, 3.2 Hz, NCH), 7.22–7.43 (m, 8H, ArCH), 7.49–7.54 (m, 2H, ArCH); ¹³C NMR (75 MHz, CDCl₃) δ 21.5 (CH₃), 22.9, 23.4, 25.1 (cyclohexyl, CHCH₂CH), 29.1 (CHCH₃), 33.2, 34.4, 35.7 (cyclohexyl), 41.2 (COCH₂), 65.4 (NCH), 86.2 (OC), 96.5 (OCN), 126.0, 127.2, 127.4, 127.5, 127.7, 128.1 (ArCH), 143.0, 146.1 (ArC), 167.8 (C=O). Anal. Calcd for C₂₅H₂₉NO₂: C, 79.96; H, 7.78; N, 3.73. Found: C, 80.01; H, 7.77; N, 3.65.

4.6. Synthesis of (–)-[(2*S*,4*R*)-1-cyclohexyl-4-methylpiperidin-2-yl]methanol, (–)-5a

To a cold (0 °C) suspension of LiAlH₄ (44 mg, 1.16 mmol) in dry THF (1.2 mL) under an argon atmosphere was added dropwise a solution of AlCl₃ (52 mg, 0.387 mmol) in dry THF (3 mL). The solution was stirred for 25 min at rt, then it was cooled to –78 °C and a solution of amido acetal **4a** (95 mg, 0.425 mmol) in dry THF (3.6 mL) was added. The solution was stirred at –78 °C for 45 min, the temperature raised to 0 °C and stirred for another 35 min. Then, the mixture was carefully hydrolysed with 1 M HCl until hydrogen evolution stopped and extracted with CHCl₃/*i*-PrOH, 80:20. The organic phase was washed with 1 M NaOH and then with brine, dried over anhydrous MgSO₄, filtered and concentrated under reduced pressure to obtain 63 mg of pure product (–)-**5a** (70% yield). White solid; mp 151–153 °C; [α]_D = –27.8 (c 0.6, CHCl₃); IR (ATR) 3256, 2932, 1452, 1101 cm^{–1}; ¹H NMR (300 MHz, CDCl₃) δ 0.95 (d, 3H, ³J = 6.3 Hz, CH₃), 1.03–1.14 (m, 1H, cyclohexyl), 1.16–1.55 (m, 6H, NCH₂CH₂, CHCH₃, cyclohexyl), 1.56–1.70 (m, 4H, NCH₂CH₂, NCHCH₂, cyclohexyl), 1.79–1.86 (m, 2H, cyclohexyl), 1.94 (d, 1H, ²J = 11.4 Hz, cyclohexyl), 2.39 (td, 1H, ²J = 11.8 Hz, ³J = 2.6 Hz, NCH₂), 2.78 (dc, 1H, ³J = 11.4, 3.2 Hz, NCHCH₂O), 3.14 (dt, 1H, ²J = 11.8 Hz, ³J = 3.3 Hz, NCH₂), 3.11–3.19 (m, 1H, NCH), 3.55 (d, 1H, ²J = 11.5 Hz, CH₂OH), 3.92 (ddd, 1H, ²J = 11.5 Hz, ³J = 5.4, 3.2 Hz, CH₂OH); ¹³C NMR (75 MHz, CDCl₃) δ 21.7 (CH₃), 24.0, 25.6, 26.1, 26.2 (cyclohexyl), 30.6 (CHCH₃), 31.0 (cyclohexyl), 33.2 (NCH₂CH₂), 37.4 (NCHCH₂), 45.4 (NCH₂), 58.0 (NCH), 59.1 (NCHCH₂O), 62.4 (CH₂OH). Anal. Calcd for C₁₃H₂₅NO: C, 73.88; H, 11.92; N, 6.63. Found: C, 73.97; H, 12.00; N, 6.56. HRMS (ESI): calcd for C₁₃H₂₅NO 211.1936; found 211.1942.

4.7. Synthesis of (–)-[(2*S*,4*R*)-1-cyclohexyl-4-methylpiperidin-2-yl](diphenyl)methanol, (–)-5b

From **4b** (38 mg, 0.105 mmol) in dry THF (1.2 mL) following the procedure described for **5a**. The reaction mixture was left at rt for 16 h, and then it was carefully hydrolysed with 1 M HCl until the hydrogen evolution stopped and was extracted with CHCl₃/*i*-PrOH, 80:20. The organic phase was washed with 1 M NaOH and then with brine, dried over anhydrous MgSO₄, filtered and concentrated under reduced pressure to obtain 22 mg of (–)-**5b** (58% yield) as a single isomer. Waxy oil; [α]_D = –25.1 (c 0.4, CH₂Cl₂); IR (ATR) 3257, 1450, 1033, 702 cm^{–1}; ¹H NMR (300 MHz, CDCl₃) δ 0.75 (d, 3H, ³J = 6.4 Hz, CH₃), 0.78–1.65 (m, 14H, CH₂CHCH₂, cyclohexyl), 2.40 (dt ap., 1H, ²J = 11.1 Hz, ³J = 3.4 Hz, cyclohexyl), 2.48 (dt, 1H, ²J = 11.7 Hz, ³J = 3.2 Hz, NCH₂), 2.93 (dt, 1H, ²J = 11.7 Hz, ³J = 4.2 Hz, NCH₂), 3.65 (t, 1H, ³J = 6.6 Hz, NCH), 3.74 (dd, 1H, ³J = 11.4, 3.8 Hz, NCHC), 7.05–7.32 (m, 6H, ArCH), 7.49–7.55 (m, 2H, ArCH), 7.69–7.75 (m, 2H, ArCH); ¹³C NMR (75 MHz, CDCl₃) δ 22.4 (CH₃), 25.2, 25.4, 26.0, 26.5 (cyclohexyl, CH₂), 30.4 (CH₃CH), 32.3, 33.8, 37.3 (cyclohexyl, CH₂), 44.5 (NCH₂), 57.5, 61.9 (CH), 77.3 (CO), 124.4, 125.6, 125.8, 126.5, 127.7, 128.1 (ArCH), 147.0, 150.4 (ArC). Anal. Calcd for C₂₅H₃₃NO: C, 82.60; H, 9.15; N, 3.85. Found: C, 82.67; H, 9.18; N, 3.86. HRMS (ESI): calcd for C₂₅H₃₃NO 363.2562; found 363.2563.

4.8. Synthesis of ligand 5c

4.8.1. (–)-(2*S*)-1-Cyclohexylpiperidine-2-carboxylic acid, (–)-13

A mixture of L-pipecolic acid (200 mg, 1.548 mmol), cyclohexanone (120 μL, 1.548 mmol) and 10% Pd on activated charcoal (tip of spatula) in dry methanol (6.5 mL) was stirred for 24 h under a hydrogen atmosphere (1.1 atm). The crude was then filtered on a thin pad of Celite, and the solvent was removed under reduced pressure and the residue was purified by flash chromatography (silica gel, methanol) to obtain 238 mg (73% yield) of acid (–)-**13**.

White solid; mp 187–188 °C (decomp.); [α]_D = –53.4 (c 0.71, MeOH); IR (ATR) 3425, 2929, 2859, 1613, 1399 cm^{–1}; ¹H NMR (300 MHz, CDCl₃) δ 1.03–1.48 (m, 5H, CHCH₂CH₂, cyclohexyl), 1.50–1.74 (m, 2H, cyclohexyl), 1.75–1.98 (m, 5H, CHCH₂CH₂CH₂, cyclohexyl), 2.00–2.25 (m, 3H, NCH₂CH₂, cyclohexyl), 2.31–2.41 (m, 1H, CHCH₂), 2.66 (td, 1H, ²J = ³J = 12.1 Hz, ³J = 2.8 Hz, NCH₂), 3.52 (d, 1H, ²J = 12.1 Hz, NCH₂), 3.59 (dd, 1H, ³J = 11.9, 3.4 Hz, COCH), 3.69 (tt, 1H, ³J = 11.7, 3.1 Hz, NCH); ¹³C NMR (75 MHz, CDCl₃) δ 22.8, 23.8, 26.3, 26.4, 26.5, 29.6, 29.9 (CHCH₂CH₂CH₂, Cyclohexyl), 47.0 (NCH₂), 67.3, 65.4 (NCH, COCH), 173.8 (C=O). Anal. Calcd for C₁₂H₂₁NO₂: C, 68.21; H, 10.02; N, 6.63. Found: C, 68.29; H, 10.00; N, 6.57.

4.8.2. Synthesis of (–)-[(2*S*)-1-cyclohexylpiperidin-2-yl]methanol, (–)-5c

To a suspension of acid (–)-**13** (88 mg, 0.416 mmol) in dioxane (2 mL) under an argon atmosphere was added dropwise a 1.0 M solution of BH₃·THF (1.3 mL), and the mixture was stirred for 24 h and carefully hydrolysed with 6 M HCl (2 mL). Next, it was stirred for another 30 min and neutralised with NaOH until neutral–basic pH. This solution was extracted with a CHCl₃/*i*-PrOH, 80:20 mixture, dried over MgSO₄, filtered and concentrated under reduced pressure. The crude was identified as (–)-**5c** (66 mg, 80% yield). White solid; mp 59–60 °C; [α]_D = –37.6 (c 0.25, MeOH); IR (ATR) 3288, 1452, 1181 cm^{–1}; ¹H NMR (300 MHz, CDCl₃) δ 0.98–1.85 (m, 16H, CHCH₂CH₂CH₂, cyclohexyl), 2.39 (ddd, 1H, ²J = 12.2 Hz, ³J = 9.4, 2.8 Hz, NCH₂), 2.71 (dc, 1H, ³J = 8.3, 4.3 Hz, OCH₂CH), 2.81 (tt, 1H, ³J = 11.0, 2.9 Hz, NCH), 2.92 (ddd, 1H, ²J = 12.2 Hz, ³J = 5.7, 3.5 Hz, NCH₂), 3.20–3.54 (s, 1H, OH), 3.47 (dd, 1H, ²J = 10.7 Hz, ³J = 4.3 Hz, CH₂OH), 3.70 (dd, 1H, ²J = 10.7 Hz, ³J = 4.3 Hz, CH₂OH); ¹³C NMR (75 MHz, CDCl₃) δ 23.2, 25.0, 25.8, 26.2, 26.2, 26.3, 27.9, 31.9 (CHCH₂CH₂CH₂, cyclohexyl), 44.5 (NCH₂), 57.2, 58.2 (CHNCH), 61.6 (CH₂OH). Anal. Calcd for C₁₂H₂₃NO: C, 73.04; H, 11.75; N, 7.10. Found: C, 73.00; H, 11.68; N, 7.14. HRMS (ESI): calcd for C₁₂H₂₃NO 197.1780; found 197.1776.

4.9. Synthesis of ligand 5d

4.9.1. (–)-(2*S*)-1-Benzylpiperidine-2-carboxylic acid, (–)-14

To a heated (40 °C) mixture of L-pipecolic acid (100 mg, 0.774 mmol) and KOH (130 mg, 2.32 mmol) in *i*-PrOH (1.5 mL) was added dropwise (over 2 h, using a syringe pump) a solution of benzyl bromide (99 μL, 0.828 mmol) in *i*-PrOH (0.5 mL) and was stirred for another 6 h at the same temperature. The mixture was hydrolysed with concd HCl (0.16 mL approx.) until pH 5–6, extracted with CHCl₃, dried over anhydrous MgSO₄, filtered and concentrated at reduced pressure. The crude was purified by flash chromatography (silica gel, methanol) to obtain 134 mg of (–)-**14** (79% yield). Colourless oil; [α]_D = –16.2 (c 0.5, EtOH); IR (ATR) 3401, 1623, 1402, 701 cm^{–1}; ¹H NMR (300 MHz, CDCl₃) δ 1.47–1.96 (m, 5H, CHCH₂CH₂CH₂), 2.37–2.20 (m, 1H, CHCH₂), 3.02 (td, 1H, ²J = 12.0 Hz, ³J = 3.5 Hz, NCH₂), 3.41 (d, 1H, ²J = 12.0 Hz, NCH₂), 3.65 (dd, 1H, ³J = 11.4, 3.2 Hz, CH), 4.18 (d, 1H, ²J = 12.6 Hz, NCH₂Ph), 4.61 (d, 1H, ²J = 12.6 Hz, NCH₂Ph), 7.61–7.40 (m, 5H, ArCH); ¹³C NMR (75 MHz, CDCl₃) δ 22.3, 22.6, 29.6 (CHCH₂CH₂CH₂), 58.3 (NCH₂), 70.0 (NCH₂Ph), 70.9 (CH), 129.1, 129.9, 131.8, 133.2, (ArCH, ArC), 170.5 (C=O). Anal. Calcd for C₁₃H₁₇NO₂: C, 71.21; H, 7.81; N, 6.39. Found: C, 71.15; H, 7.73; N, 6.43.

4.9.2. Synthesis of (–)-[(2*S*)-1-benzylpiperidin-2-yl]methanol, (–)-5d

To a cold (0 °C) suspension of (–)-**14** (94 mg, 0.429 mmol) in dry diethyl ether (2 mL) under an argon atmosphere was cannuled a solution of LiAlH₄ (65 mg, 1.715 mmol) in dry diethyl ether (1 mL), and the mixture was stirred for 1 h at the same temperature and then overnight at rt. Then, the mixture was cooled to

0 °C and was carefully hydrolysed with a saturated solution of Na₂SO₄ (2 mL), extracted with a CHCl₃/i-PrOH, 80:20 mixture, dried over MgSO₄, filtered and concentrated under reduced pressure to obtain 75 mg of (–)-**5d** (85% yield). Colourless oil; [α]_D = –31.4 (c 0.5, CHCl₃); IR (ATR) 3393, 1452, 1061, 734, 698 cm^{–1}; ¹H NMR (300 MHz, CDCl₃) δ 1.14–1.73 (m, 6H, CHCH₂CH₂CH₂), 2.11 (ddd, 1H, ²J = 12.3 Hz, ³J = 10.3, 3.1 Hz, NCH₂), 2.42 (dc, 1H, ³J = 8.5, 4.1 Hz, CH), 2.76 (s, 1H, OH), 2.83 (dddd, 1H, ²J = 12.3 Hz, ³J = 5.1, 3.7 Hz, ⁴J = 1.3 Hz, NCH₂), 3.29 (d, 1H, ²J = 13.4 Hz, NCH₂Ph), 3.49 (ABX system, 1H, ²J = 10.9 Hz, ³J = 4.1 Hz, CH₂OH), 3.83 (ABX system, 1H, ²J = 10.9 Hz, ³J = 4.1 Hz, CH₂OH), 4.03 (d, 1H, ²J = 13.4 Hz, NCH₂Ph), 7.17–7.34 (m, 5H, ArCH); ¹³C NMR (75 MHz, CDCl₃) δ 23.3, 24.0, 27.2 (CHCH₂CH₂CH₂), 50.8 (NCH₂), 57.6 (NCH₂Ph), 60.9 (CH), 62.2 (CH₂OH), 127.0, 128.3, 128.8 (ArCH), 138.9 (ArC). Anal. Calcd for C₁₃H₁₉NO: C, 76.06; H, 9.33; N, 6.82. Found: C, 76.21; H, 9.41; N, 6.78. HRMS (ESI): calcd for C₁₃H₁₉NO 205.1467; found 205.1471.

4.10. Synthesis of ligand **5e**

4.10.1. Synthesis of (–)-(2S)-1-methylpiperidine-2-carboxylic acid, (–)-**15**

A mixture of L-pipecolic acid (100 mg, 0.774 mmol), paraformaldehyde (47 mg, 1.55 mmol) and 10% Pd on activated charcoal (tip of spatula) in dry methanol (3.5 mL) was stirred for 24 h under a hydrogen atmosphere (1.1 atm). The crude was then filtered on a thin pad of Celite, and the solvent was removed under reduced pressure. The residue was purified by flash chromatography (silica gel, methanol) to obtain 111 mg (ca. quantitative yield) of acid (–)-**15**. White solid; mp 213–214 °C (decomp.); [α]_D = –62.2 (c 0.8, MeOH); IR (ATR) 3425, 2942, 1620, 1320, 1179 cm^{–1}; ¹H NMR (300 MHz, CDCl₃) δ 1.46–1.61 (m, 1H, CHCH₂CH₂), 1.65–1.91 (m, 4H, CHCH₂CH₂CH₂), 2.15–2.24 (m, 1H, CHCH₂), 2.81 (s, 3H, CH₃), 2.91 (td, 1H, ²J = 12.3 Hz, ³J = 3.3 Hz, NCH₂), 3.28 (dd, 1H, ³J = 11.4, 3.5 Hz, CH), 3.39 (dt, 1H, ²J = 12.3 Hz, ³J = 4.0 Hz, NCH₂); ¹³C NMR (75 MHz, CDCl₃) δ 23.1 (CHCH₂CH₂), 24.5 (NCH₂CH₂), 29.8 (CHCH₂), 43.4 (CH₃), 55.5 (NCH₂), 70.7 (CH), 174.4 (C=O). Anal. Calcd for C₇H₁₃NO₂: C, 58.72; H, 9.15; N, 9.78. Found: C, 58.81; H, 9.24; N, 9.73.

4.10.2. Synthesis of (+)-[(2S)-1-methylpiperidin-2-yl]methanol, (+)-**5e**

To a suspension of acid (–)-**15** (50 mg, 0.349 mmol) in dioxane (1 mL) under an argon atmosphere was added dropwise a 1.0 M solution of BH₃·THF (1.1 mL), and the mixture was stirred for 24 h and carefully hydrolysed with 6 M HCl (1 mL). The mixture was stirred for another 30 min and neutralised with NaOH until neutral–basic pH. This solution was extracted with a CHCl₃/i-PrOH, 80:20 mixture, dried over MgSO₄, filtered and concentrated under reduced pressure to obtain (+)-**5e** (40 mg, 89% yield). Waxy solid; [α]_D = +3.9 (c 0.25, MeOH); IR (ATR) 3322, 1462, 1060 cm^{–1}; ¹H NMR (300 MHz, CDCl₃) δ 1.22–1.43 (m, 1H, NCH₂CH₂), 1.61–1.85 (m, 4H, CHCH₂CH₂), 2.24–2.41 (m, 2H, NCH₂CH₂), 2.47 (s, 3H, CH₃), 3.05 (dtd, 1H, ²J = 11.9 Hz, ³J = 3.5 Hz, ⁴J = 1.2 Hz, NCH₂), 3.53 (dd, 1H, ²J = 11.5 Hz, ³J = 3.0 Hz, CH₂OH), 3.60–3.77 (m, 1H, CH), 3.80 (dd, 1H, ²J = 11.5 Hz, ³J = 3.7 Hz, CH₂OH), 4.18 (s, 1H, OH); ¹³C NMR (75 MHz, CDCl₃) δ 23.4, 24.4, 27.5 (CHCH₂CH₂CH₂), 42.0 (CH₃), 56.4 (NCH₂), 62.5 (CH), 64.7 (CH₂OH). Anal. Calcd for C₇H₁₅NO: C, 65.07; H, 11.70; N, 10.84. Found: C, 65.00; H, 11.62; N, 10.81. HRMS (ESI): calcd for C₇H₁₅NO 129.1154; found 129.1156.

4.11. Synthesis of ligand **5f**

4.11.1. Ethyl (–)-(2S)-piperidine-2-carboxylate, (–)-**16**

A solution of L-pipecolic acid (100 mg, 0.774 mmol) in freshly distilled SOCl₂ (2 mL) was stirred for 16 h at rt. Then, the excess

SOCl₂ was removed by evaporation under reduced pressure and the residue was dissolved in absolute ethanol (3 mL) and stirred for another 6 h. The solvent was removed under reduced pressure yielding 144 mg (95% yield) of (–)-**16**. Colourless oil; [α]_D = –11.8 (c 0.28, CHCl₃); ¹H NMR (300 MHz, CDCl₃) δ 1.28 (t, 3H, ³J = 7.2 Hz, CH₃), 1.40–1.62 (m, 4H, CHCH₂CH₂CH₂), 1.74–1.84 (m, 1H, NHCH₂CH₂), 1.97 (dc, 1H, ²J = 12.7 Hz, ³J = 3.2 Hz, CHCH₂), 2.06 (s, 1H, NH), 2.67 (ddd, 1H, ²J = 12.2 Hz, ³J = 10.3, 3.2 Hz, NCH₂), 3.10 (dtd, 1H, ²J = 12.2 Hz, ³J = 3.7, 1.2 Hz, NCH₂), 3.35 (dd, 1H, ³J = 9.7, 3.2 Hz, CH), 4.18 (c, 2H, ³J = 7.2 Hz, OCH₂); ¹³C NMR (75 MHz, CDCl₃) δ 14.3 (CH₃), 24.0, 25.8, 29.1 (CHCH₂CH₂CH₂), 45.7 (NCH₂), 58.6 (CH), 60.6 (OCH₂), 173.4 (C=O).

4.11.2. (–)-Diphenyl-[(2S)-piperidin-2-yl]methanol, (–)-**17**

To a cold (0 °C) solution of (–)-**16** (143 mg, 0.783 mmol) in dry THF (8 mL), under an argon atmosphere, was added a solution of PhMgBr (2.3 mL, 6.64 mmol). The mixture was stirred for 16 h at rt, and then was diluted with diethyl ether (100 mL) and carefully hydrolysed with a saturated aqueous NH₄Cl solution (25 mL). The organic layer was washed with brine (25 mL), dried over anhydrous MgSO₄, filtered and concentrated under reduced pressure. The crude was purified by flash chromatography (silica gel; DCM and MeOH) to obtain 144 mg (73% yield) of the alcohol (–)-**17**. White solid; mp 74–75 °C; [α]_D = –75.7 (c 0.75, CHCl₃); IR (ATR) 3297, 3061, 2941, 1588, 1449, 745 cm^{–1}; ¹H NMR (300 MHz, CDCl₃) δ 1.28–1.41 (m, 2H, CHCH₂CH₂), 1.42–1.69 (m, 3H, NCH₂CH₂, CHCH₂), 1.80 (dt, 1H, ²J = 12.8 Hz, ³J = 3.8 Hz, CHCH₂CH₂), 2.78 (dt, 1H, ²J = 11.8 Hz, ³J = 3.0 Hz, NCH₂), 3.14 (d, 1H, ²J = 11.8 Hz, NCH₂), 3.65 (dd, 1H, ³J = 11.7, 2.5 Hz, CH), 5.34 (s, 2H, NH, OH), 7.11–7.67 (m, 10H, ArCH); ¹³C NMR (75 MHz, CDCl₃) δ 24.1, 24.6, 24.9 (CHCH₂CH₂CH₂), 46.6 (NCH₂), 62.7 (CH), 78.3 (COH), 125.4, 125.8, 126.5, 127.1, 128.0, 128.7 (ArCH), 144.0, 145.4 (ArC). Anal. Calcd for C₁₈H₂₁NO: C, 80.86; H, 7.92; N, 5.24. Found: C, 80.91; H, 7.99; N, 5.20.

4.11.3. (8aS)-1,1-Diphenylhexahydro[1,3]oxazolo[3,4-a]pyridine, **18**

A mixture of (–)-**17** (112 mg, 0.419 mmol) and paraformaldehyde (111 mg, 3.686 mmol) in MeOH (4.4 mL) was stirred for 3 h. Next, the mixture was cooled to 0 °C and NaBH₄ (72 mg, 1.885 mmol) was added. The mixture was stirred for another 16 h at rt and then was concentrated under reduced pressure. The crude was dissolved in water (15 mL) and extracted with ethyl acetate (2–15 mL). The combined organic extracts were dried over anhydrous MgSO₄, filtered and concentrated under reduced pressure. The crude was identified as the intermediate oxazolidine **18** (67 mg, 57% yield) and was directly transformed into the alcohol. Colourless oil. ¹H NMR (300 MHz, CDCl₃) δ 0.83–0.91 (m, 1H, CHCH₂), 1.41–1.51 (m, 2H, NCH₂CH₂), 1.60–1.69 (m, 1H, CHCH₂CH₂), 1.75–1.84 (m, 1H, CHCH₂CH₂), 1.87–1.96 (m, 1H, CHCH₂), 2.13 (ddd, 1H, ²J = 10.6 Hz, ³J = 11.7, 3.1 Hz, NCH₂), 2.82 (dd, 1H, ³J = 11.0 Hz, 2.3 Hz, CH), 3.14 (dt, 1H, ²J = 10.6 Hz, ³J = 3.8 Hz, NCH₂), 4.02 (d, 1H, ²J = 1.5 Hz, OCH₂N), 4.99 (d, 1H, ²J = 1.5 Hz, OCH₂N), 7.19–7.51 (m, 10H, ArCH); ¹³C NMR (75 MHz, CDCl₃) δ 23.97, 24.02 (NCH₂CH₂CH₂), 28.1 (CHCH₂), 46.7 (NCH₂), 70.4 (CH), 85.9 (OCH₂N), 88.2 (OC), 126.6, 126.7, 127.1, 127.2, 127.4, 128.0 (ArCH), 143.9, 145.4 (ArC).

4.11.4. Synthesis of (–)-(2S)-1-methylpiperidin-2-yl (diphenyl)methanol, (–)-**5f**

To a cold (0 °C) suspension of LiAlH₄ (28 mg, 0.783 mmol) in dry THF (2 mL) under an argon atmosphere was added dropwise a solution of AlCl₃ (32 mg, 0.24 mmol) in dry THF (2.5 mL). The solution was stirred for 30 min at rt, then cooled to –78 °C and a solution of **18** (67 mg, 0.24 mmol) in dry THF (2 mL) was added. The solution was stirred at –78 °C for 45 min. The temperature was increased to 0 °C and the mixture was carefully hydrolysed with 1 M

HCl until hydrogen evolution ceased. The resulting suspension was extracted with $\text{CHCl}_3/i\text{-PrOH}$, 80:20 ($3 \times 10\text{ mL}$) and the combined organic extracts were washed with 1 M NaOH (10 mL) and then with brine (10 mL), dried over anhydrous MgSO_4 , filtered and concentrated under reduced pressure to obtain 65 mg of pure product (–)-**5f** (97% yield). Colourless oil; $[\alpha]_D^{25} = -75.8$ ($c\ 1.0$, CHCl_3); IR (ATR) 3349, 1449, 1030, 706 cm^{-1} ; ^1H NMR (300 MHz, CDCl_3) δ 1.21–1.35 (m, 2H, piperidine), 1.50–1.71 (m, 4H, piperidine), 2.02 (s, 3H, CH_3), 2.47 (td, 1H, $^2J = ^3J = 11.5\text{ Hz}$, $^3J = 4.0\text{ Hz}$, NCH_2), 2.96 (ddd, 1H, $^2J = 11.5\text{ Hz}$, $^3J = 5.3$, 3.5 Hz, NCH_2), 3.36 (dd, 1H, $^3J = 9.7$, 4.3 Hz, CH), 7.06–7.76 (m, 10H, ArCH); ^{13}C NMR (75 MHz, CDCl_3) δ 23.9 (CHCH_2CH_2), 24.7 (NCH_2CH_2), 27.9 (CHCH_2), 46.4 (CH_3), 57.6 (NCH_2), 67.7 (CH), 77.4 (CO), 124.9, 125.7, 125.8, 126.2, 127.9, 128.1 (ArCH), 146.6, 149.8 (ArC). Anal. Calcd for $\text{C}_{19}\text{H}_{23}\text{NO}$: C, 81.10; H, 8.24; N, 4.98. Found: C, 81.21; H, 8.34; N, 4.79. HRMS (ESI): calcd for $\text{C}_{19}\text{H}_{23}\text{NO}$ 281.1780; found 281.1787.

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